



Buckets & Data Governance

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Strategic Storage Management for OpenPipeline Logs

This notebook covers Grail bucket architecture, retention policies, routing configuration, access control, and cost optimization strategies.

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Prerequisites

-  Access to a Dynatrace environment with log data

- ☒ Understanding of OpenPipeline basics (OPLOGS-01 to OPLOGS-03)
- ☒ Admin access for bucket configuration (optional)

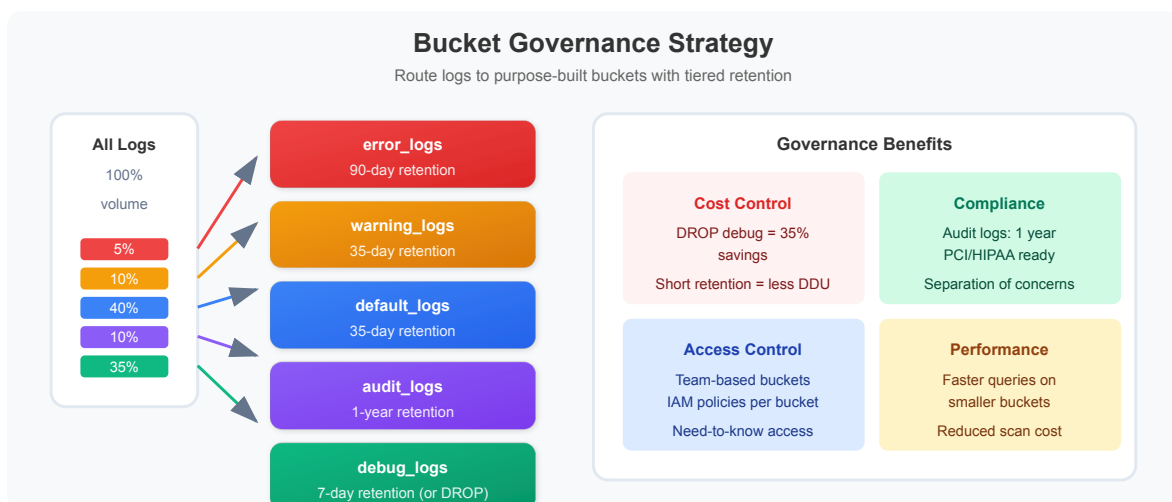
```
// Discover all buckets with log data
fetch logs, from: now() - 24h
| summarize {log_count = count()}, by: {dt.system.bucket}
| sort log_count desc
```

```
// Bucket volume analysis over time
fetch logs, from: now() - 7d
| makeTimeseries {count = count()}, by: {dt.system.bucket}, interval: 1d
| limit 100
```

```
// Bucket composition by log level
fetch logs, from: now() - 24h
| summarize {count = count()}, by: {dt.system.bucket, loglevel}
| sort dt.system.bucket asc, count desc
```

2. Strategic Bucket Design

Design buckets based on **access patterns**, **retention requirements**, and **cost optimization**.



Recommended Bucket Strategy

Bucket Name	Purpose	Retention	Access	Data Types
default_logs	General purpose	35 days	All teams	Standard logs
debug_logs	Development troubleshooting	7 days	Dev teams	DEBUG, TRACE
error_logs	Incident investigation	90 days	SRE/Ops	ERROR, FATAL
audit_logs	Compliance	365 days	Audit team	Auth, access
security_logs	Security operations	180 days	SecOps	Security events
archive_logs	Long-term storage	730 days	Compliance	Critical events

Bucket Naming Convention

```
<purpose>_logs
<team>_<purpose>_logs
<environment>_<purpose>_logs
```

Examples:

- platform_debug_logs
- prod_security_logs
- payment_audit_logs

```
// Analyze current log distribution for bucket planning
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    debug = countIf(loglevel == "DEBUG" OR loglevel == "TRACE"),
    info = countIf(loglevel == "INFO" OR loglevel == "NOTICE"),
    warn = countIf(loglevel == "WARN" OR loglevel == "WARNING"),
    error = countIf(loglevel == "ERROR" OR loglevel == "FATAL")
}
| fieldsAdd debug_pct = round((debug * 100.0) / total, decimals: 1)
| fieldsAdd info_pct = round((info * 100.0) / total, decimals: 1)
| fieldsAdd warn_pct = round((warn * 100.0) / total, decimals: 1)
| fieldsAdd error_pct = round((error * 100.0) / total, decimals: 1)
```

```
// Identify audit/security logs for specialized bucket
fetch logs, from: now() - 24h
| filter contains(content, "auth")
    OR contains(content, "login")
    OR contains(content, "logout")
    OR contains(content, "permission")
    OR contains(content, "access")
| summarize {count = count()}, by: {k8s.namespace.name, loglevel}
| sort count desc
| limit 20
```

```
// Volume by source - helps determine routing needs
fetch logs, from: now() - 24h
| summarize {count = count()}, by: {dt.openpipeline.source, dt.system.bucket}
| sort count desc
```

3. Retention Policies

Retention Strategy Matrix

Data Type	Min Retention	Recommended	Max	Rationale
DEBUG/TRACE	3 days	7 days	14 days	High volume, temporary
INFO	7 days	35 days	90 days	Standard operations
WARN	14 days	60 days	180 days	Trend analysis
ERROR	30 days	90 days	365 days	Incident investigation
Audit	365 days	365 days	730 days	Compliance
Security	90 days	180 days	365 days	SIEM/forensics

Retention Cost Impact

Storage Cost = Log Volume × Retention Days × Cost per GB-day

Example:

- 100 GB/day DEBUG logs × 35 days = 3,500 GB stored
- 100 GB/day DEBUG logs × 7 days = 700 GB stored
- Savings: 80% reduction in DEBUG storage

```
// Estimate storage by log level
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
  log_count = count(),
  total_mb = round(sum(content_bytes) / 1048576.0, decimals: 2),
  avg_bytes = round(avg(content_bytes), decimals: 0)
}, by: {loglevel}
| sort total_mb desc
```

```
// Calculate retention cost comparison (DEBUG logs)
fetch logs, from: now() - 24h
| filter loglevel == "DEBUG" OR loglevel == "TRACE"
| fieldsAdd content_bytes = stringLength(content)
| summarize {daily_mb = round(sum(content_bytes) / 1048576.0, decimals: 2)}
| fieldsAdd retention_35d_gb = round(daily_mb * 35 / 1024, decimals: 2)
| fieldsAdd retention_7d_gb = round(daily_mb * 7 / 1024, decimals: 2)
| fieldsAdd savings_gb = round(retention_35d_gb - retention_7d_gb, decimals:
2)
| fieldsAdd savings_pct = round((savings_gb * 100) / retention_35d_gb,
decimals: 1)
```

```
// Retention planning: volume by bucket and loglevel
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    log_count = count(),
    daily_mb = round(sum(content_bytes) / 1048576.0, decimals: 2)
}, by: {dt.system.bucket, loglevel}
| sort dt.system.bucket asc, daily_mb desc
```

4. Bucket Routing Configuration

Route logs to appropriate buckets using **OpenPipeline routing rules**.

OpenPipeline Routing Rules

```
# Route logs based on conditions
routes:
  # Route debug logs to short-retention bucket
  - name: debug-to-debug-bucket
    condition: loglevel == "DEBUG" OR loglevel == "TRACE"
    bucket: debug_logs

  # Route errors to extended-retention bucket
  - name: errors-to-error-bucket
    condition: loglevel == "ERROR" OR loglevel == "FATAL"
    bucket: error_logs

  # Route audit logs to compliance bucket
  - name: audit-to-audit-bucket
    condition: contains(content, "audit") OR contains(k8s.namespace.name,
"security")
    bucket: audit_logs

  # Default catch-all
  - name: default-route
    condition: true
    bucket: default_logs
```

Routing Priority

1. First matching rule wins
2. Order rules from specific to general
3. Always include a default catch-all

```
// Preview routing decision logic
fetch logs, from: now() - 1h
| limit 1000
| fieldsAdd target_bucket = if(loglevel == "DEBUG" OR loglevel == "TRACE",
"debug_logs",
                                else: if(loglevel == "ERROR" OR loglevel ==
"FATAL", "error_logs",
                                else: if(contains(content, "audit") OR
contains(content, "security"), "audit_logs",
                                else: "default_logs"))
| summarize {count = count()}, by: {target_bucket, loglevel}
| sort target_bucket asc, count desc
```

```
// Validate current routing effectiveness
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    in_default = countIf(dt.system.bucket == "default_logs")
}, by: {loglevel}
| fieldsAdd default_pct = round((in_default * 100.0) / total, decimals: 1)
| sort default_pct desc
```

```
// Namespace-based routing analysis
fetch logs, from: now() - 24h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd suggested_bucket = if(contains(k8s.namespace.name, "prod"),
"prod_logs",
                                else: if(contains(k8s.namespace.name,
"staging"), "staging_logs",
                                else: "dev_logs"))
| summarize {count = count()}, by: {k8s.namespace.name, suggested_bucket}
| sort count desc
| limit 20
```

5. Access Control & Data Governance

Bucket-Level Access Control

Bucket	Read Access	Write Access	Use Case
default_logs	All authenticated	OpenPipeline	General access
debug_logs	Dev teams	OpenPipeline	Development
audit_logs	Audit team only	OpenPipeline	Compliance
security_logs	SecOps only	OpenPipeline	Security ops

IAM Policy Integration

Configure in **Settings → Access tokens & IAM → Policies**


```
{
  "name": "audit-log-readers",
  "statement": [
    {
      "effect": "ALLOW",
      "permissions": ["storage:logs:read"],
      "conditions": [
        {
          "bucket": ["audit_logs"]
        }
      ]
    }
  ]
}
```

```
// Identify sensitive data patterns for access control planning
fetch logs, from: now() - 24h
| filter contains(content, "password")
      OR contains(content, "secret")
      OR contains(content, "token")
      OR contains(content, "api_key")
      OR contains(content, "credential")
| summarize {count = count()}, by: {dt.system.bucket, k8s.namespace.name}
| sort count desc
| limit 15
```

```
// Data governance audit: bucket access patterns
fetch logs, from: now() - 24h
| summarize {
  total_logs = count(),
  unique_sources = countDistinct(dt.openpipeline.source)
}, by: {dt.system.bucket}
| sort total_logs desc
```

6. Query Optimization with Buckets

Target Specific Buckets

Querying specific buckets **reduces scan costs** and **improves performance**.

✅ **Efficient - targets specific bucket:**

```
fetch logs, bucket: "error_logs", from: now() - 7d
| filter loglevel == "ERROR"
| limit 100
```

❌ Inefficient - scans all buckets:

```
fetch logs, from: now() - 7d
| filter loglevel == "ERROR"
| limit 100
```

Multi-Bucket Queries

```
// Query multiple specific buckets
fetch logs, bucket: {"prod_logs", "error_logs"}, from: now() - 24h
| filter loglevel == "ERROR"
```

```
// Query specific bucket (efficient)
fetch logs, bucket: "default_logs", from: now() - 1h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

```
// Compare bucket query efficiency
fetch logs, from: now() - 1h
| summarize {
  total_logs = count(),
  buckets_scanned = countDistinct(dt.system.bucket)
}
```

```
// Bucket-aware dashboarding: hourly volume by bucket
fetch logs, from: now() - 24h
| makeTimeseries {count = count()}, by: {dt.system.bucket}, interval: 1h
| limit 100
```

7. Cost Optimization Strategies

Storage Cost Reduction Tactics

Strategy	Savings	Implementation
Shorten DEBUG retention	60-80%	Route to 7-day bucket
Drop health check logs	10-30%	OpenPipeline filter
Sample verbose logs	50-90%	Sampling processor
Compress long content	20-40%	Field truncation

ROI Calculation

Current State:

- 100 GB/day total logs
- 35-day retention all buckets
- Total storage: 3,500 GB

Optimized State:

- 60 GB/day INFO/WARN (35d) = 2,100 GB
- 30 GB/day DEBUG (7d) = 210 GB
- 10 GB/day ERROR (90d) = 900 GB
- Total storage: 3,210 GB

Savings: ~8% immediate, scales with volume

```
// Calculate potential cost savings
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    daily_gb = round(sum(content_bytes) / 1073741824.0, decimals: 2),
    log_count = count()
}, by: {loglevel}
| fieldsAdd current_35d_gb = round(daily_gb * 35, decimals: 1)
| fieldsAdd optimal_retention = if(loglevel == "DEBUG" OR loglevel ==
"TRACE", 7,
                                else: if(loglevel == "ERROR" OR loglevel ==
"FATAL", 90,
                                else: 35))
| fieldsAdd optimal_storage_gb = round(daily_gb * optimal_retention,
decimals: 1)
| fieldsAdd savings_gb = round(current_35d_gb - optimal_storage_gb, decimals:
1)
| sort savings_gb desc
```

```
// Identify high-volume low-value logs
fetch logs, from: now() - 24h
| filter loglevel == "DEBUG" OR loglevel == "INFO"
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    log_count = count(),
    total_mb = round(sum(content_bytes) / 1048576.0, decimals: 2)
}, by: {k8s.namespace.name, k8s.workload.name}
| sort total_mb desc
| limit 20
```

8. Bucket Management Best Practices

Do's

1. **Start with clear bucket strategy** before configuring pipelines
2. **Name buckets descriptively** with purpose and team
3. **Set retention based on data value**, not convenience
4. **Review bucket distribution** weekly
5. **Query specific buckets** for performance
6. **Document bucket purposes** and access policies

❌ Don'ts

1. **Don't put everything in default bucket** - defeats purpose
2. **Don't set uniform retention** - wastes storage on low-value data
3. **Don't ignore bucket metrics** - monitor distribution
4. **Don't create too many buckets** - complexity overhead
5. **Don't skip access controls** - security risk

```
// Weekly bucket health check query
fetch logs, from: now() - 7d
| summarize {
    total_logs = count(),
    unique_sources = countDistinct(dt.openpipeline.source),
    debug_pct = round((countIf(loglevel == "DEBUG") * 100.0) / count(),
decimals: 1),
    error_pct = round((countIf(loglevel == "ERROR") * 100.0) / count(),
decimals: 1)
}, by: {dt.system.bucket}
| sort total_logs desc
```

```
// Bucket routing effectiveness
fetch logs, from: now() - 24h
| summarize {count = count()}, by: {dt.system.bucket,
dt.openpipeline.pipelines}
| sort count desc
```



Summary

In this notebook, you learned:

- ✅ **Bucket Architecture** - How Grail organizes log storage
- ✅ **Strategic Design** - Creating purpose-driven buckets
- ✅ **Retention Policies** - Optimizing data lifecycle costs
- ✅ **Routing Rules** - Directing logs to appropriate storage
- ✅ **Access Control** - Implementing data governance
- ✅ **Query Optimization** - Targeting buckets for efficiency

Key Takeaway

Buckets are foundational. Design your bucket strategy first - it impacts security, cost, and every query you'll write.

Next Steps

Continue to **OPLOGS-05: Querying & Parsing** to learn advanced DQL techniques for log analysis.

References

- [Grail Buckets](#)
- [Log Storage Configuration](#)
- [Data Retention](#)
- [IAM Policies](#)

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